LOCAL LEFSCHETZ FULL FAITHFULNESS FOR THE ETALE PUNCTURED NEIGHBOURHOOD UNDER A DUALIZING COMPLEX, HENSELIAN PAIR, AND EITHER AN S-2 PLUS THREE-DIMENSIONAL-COMPONENT CONDITION OR A DEPTH-VERSUS-DIMENSION INEQUALITY STRICTLY ABOVE TWO.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4816-4834 of sources/stacks-pione/pinned-original-source.tex; proof span is lines 4836-4860, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-pione wave 34d.

Editorial note. rqual sv-mismatch fix 2026-10-14: source_version aligned to SOURCE.json (semicolon+git form, intake authority; package evidence.json already correct); revision bumped for re-assembly per publication rotation (takeover-line defer ticket, relayed by the s15 line).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 0339: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EKT: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A80: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BJM: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BLI: sources/stacks-pione/pinned-original-source.tex, source lines 4733–4741. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EL8: sources/stacks-pione/pinned-original-source.tex, source lines 4175–4189. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EHW: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AW6: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BIR: sources/stacks-pione/pinned-original-source.tex, source lines 4860–4860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BLF: sources/stacks-pione/pinned-original-source.tex, source lines 4651–4657. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0BLF. Assume*

(a) *A has a dualizing complex,*
(b) *the pair $(A, (f))$ is henselian,*
(c) *one of the following is true*
  (i) *A_f is (S_2) and every irreducible component of X not contained in X_0 has dimension ≥ 3, or*
  (ii) *for every prime $\mathfrak{p} \subset A$, $f \notin \mathfrak{p}$ we have $depth(A_\mathfrak{p}) + \dim(A/\mathfrak{p}) > 2$.*

Then the restriction functor $F\acute{E}t_U \longrightarrow F\acute{E}t_{U_0}$ is fully faithful.

Proof. Let A' be the $\mathfrak{m}$-adic completion of A. We will show that the hypotheses remain true for A'. This is clear for conditions (a) and (b). Condition (c)(ii) is preserved by Local Cohomology, Lemma 0EHW. Next, assume (c)(i) holds. Since A is universally catenary (Dualizing Complexes, Lemma 0A80) we see that every irreducible component of $\mathrm{Spec}(A')$ not contained in $V(f)$ has dimension ≥ 3, see More on Algebra, Proposition 0AW6. Since $A \to A'$ is flat with Gorenstein fibres, the condition that A_f is (S_2) implies that A'_f is (S_2). References used: Dualizing Complexes, Section 0BJM, More on Algebra, Section 0BIR, and Algebra, Lemma 0339. Thus by Lemma 0BLI we may assume that A is a Noetherian complete local ring.

Assume A is a complete local ring in addition to the other assumptions. By Lemma 0EL8 the result follows from Algebraic and Formal Geometry, Lemma 0EKT. □